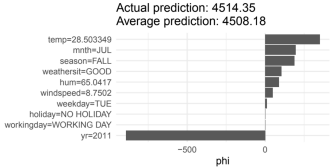
Interpretable Machine Learning

Shapley Values for Local Explanations

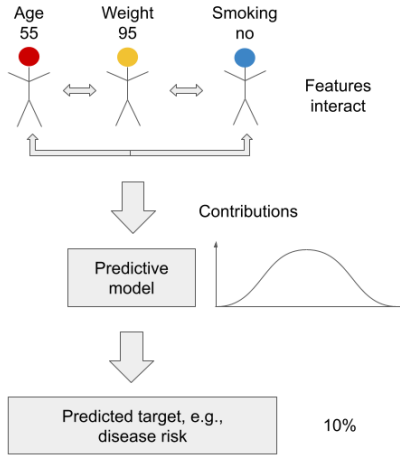


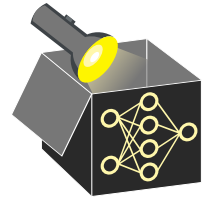
Learning goals

- See model predictions as a cooperative game
- Transfer the Shapley value concept from game theory to machine learning

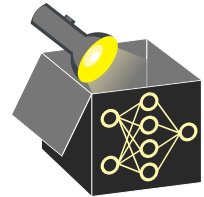
FROM GAME THEORY TO MACHINE LEARNING

- Model prediction depends on feature interactions for a specific observation
- **Goal:** Decompose prediction into **individual feature contributions**
- **Idea:** Treat features as players jointly producing a prediction
- How to fairly assign credit to features? $\Rightarrow$ Shapley values





FROM GAME THEORY TO MACHINE LEARNING

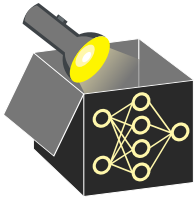


- **Game:** Predict $\hat{f}(x_1, x_2, \ldots, x_p)$ for a single observation $\mathbf{x}$
- **Players:** Features $x_j, j \in \{1, \ldots, p\}$, cooperate to produce a prediction
- **Value function:** Defines payout of coalition $S \subseteq P$ for observation $\mathbf{x}$ by

$$v(S) = \hat{f}_S(\mathbf{x}_S) - \hat{f}_\emptyset, \text{ where}$$

  - $\hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$ is the PD function $\hat{f}_S(\mathbf{x}_S) := \int \hat{f}(\mathbf{x}_S, X_{-S})\, d\mathbb{P}_{X_{-S}}$
    $\rightsquigarrow$ "Removes" features in $-S$ by marginalizing, keeping $\hat{f}$ fixed
  - Mean prediction $\hat{f}_\emptyset := \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ is subtracted to ensure $v(\emptyset) = 0$
- **Goal:** Distribute total payout $v(P) = \hat{f}(\mathbf{x}) - \hat{f}_\emptyset$ fairly among features

FROM GAME THEORY TO MACHINE LEARNING



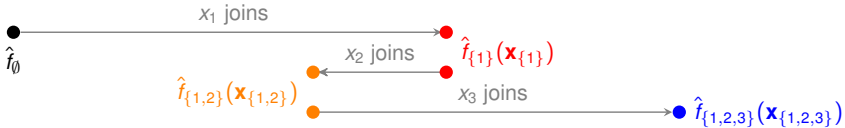
- **Game:** Predict $\hat{f}(x_1, x_2, \ldots, x_p)$ for a single observation $\mathbf{x}$
- **Players:** Features $x_j, j \in \{1, \ldots, p\}$, cooperate to produce a prediction
- **Value function:** Defines payout of coalition $S \subseteq P$ for observation $\mathbf{x}$ by

$$v(S) = \hat{f}_S(\mathbf{x}_S) - \hat{f}_\emptyset, \text{ where}$$

  - $\hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$ is the PD function $\hat{f}_S(\mathbf{x}_S) := \int \hat{f}(\mathbf{x}_S, X_{-S})\, d\mathbb{P}_{X_{-S}}$
    $\rightsquigarrow$ "Removes" features in $-S$ by marginalizing, keeping $\hat{f}$ fixed
  - Mean prediction $\hat{f}_\emptyset := \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ is subtracted to ensure $v(\emptyset) = 0$
- **Goal:** Distribute total payout $v(P) = \hat{f}(\mathbf{x}) - \hat{f}_\emptyset$ fairly among features
- **Marginal contribution of feature j joining coalition S** ($\hat{f}_\emptyset$ cancels):

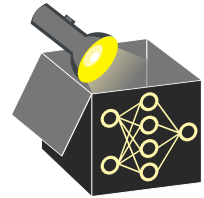
$$\Delta(j, S) = v(S \cup \{j\}) - v(S) = \hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - \hat{f}_S(\mathbf{x}_S)$$

- **Example (3 features):** Illustration how features incrementally shift $\hat{f}_S(\mathbf{x}_S)$ from baseline $\hat{f}_\emptyset$ to full prediction $\hat{f}(\mathbf{x})$ – each step reflects a marginal contribution



SHAPLEY VALUE - DEFINITION

▸ Shapley (1953) ▸ Strumbelj et al. (2014)



Order definition: Shapley value $\phi_j(\mathbf{x})$ quantifies contribution of x_j via

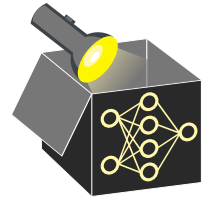
$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \underbrace{\hat{f}_{S_j^\tau \cup \{j\}}(\mathbf{x}_{S_j^\tau \cup \{j\}}) - \hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})}_{\Delta(j, S_j^\tau) \text{ marginal contribution of feature } j}$$

- **Interpretation:** $\phi_j(\mathbf{x})$ quantifies how much feature x_j contributes to the difference between $\hat{f}(\mathbf{x})$ and the mean prediction $\hat{f}_\emptyset$
  $\rightsquigarrow$ Marginal contributions and Shapley values can be negative
- **Exact computation of ϕ_j:** Using PD function $\hat{f}_S(\mathbf{x}_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S}^{(i)})$ yields

$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \frac{1}{n} \sum_{i=1}^n \hat{f}(\mathbf{x}_{S_j^\tau \cup \{j\}}, \mathbf{x}_{-\{S_j^\tau \cup \{j\}\}}^{(i)}) - \hat{f}(\mathbf{x}_{S_j^\tau}, \mathbf{x}_{-S_j^\tau}^{(i)})$$

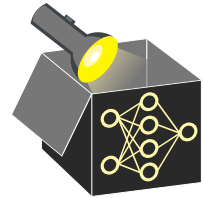
$\rightsquigarrow$ $\hat{f}_S$ marginalizes over all features not in S using all observations $i = 1, \ldots, n$
$\rightsquigarrow$ Exact computation requires $|P|! \cdot n$ marginal contribution terms

ESTIMATION: A PRACTICAL PROBLEM



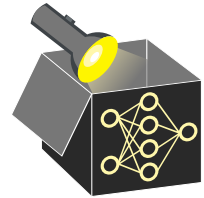
- **Exact computation is infeasible for many features:**
  For $|P| = 10$, the number of permutations is $10! \approx 3.6$ million
  $\rightsquigarrow$ Complexity grows factorially with feature count

ESTIMATION: A PRACTICAL PROBLEM



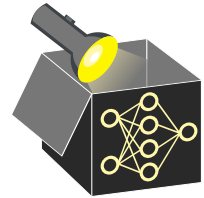
- **Exact computation is infeasible for many features:**
  For $|P| = 10$, the number of permutations is $10! \approx 3.6$ million
  $\rightsquigarrow$ Complexity grows factorially with feature count
- **Additional challenge: Estimating marginal predictions (PD functions)**
  Each permutation τ defines a coalition S_j^τ needing its own estimate of $\hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})$
  $\rightsquigarrow$ With $|P|!$ permutations and n data points, the number of such estimates grows rapidly, making marginalization costly

ESTIMATION: A PRACTICAL PROBLEM



- **Exact computation is infeasible for many features:**
  For $|P| = 10$, the number of permutations is $10! \approx 3.6$ million
  $\rightsquigarrow$ Complexity grows factorially with feature count
- **Additional challenge: Estimating marginal predictions (PD functions)**
  Each permutation τ defines a coalition S_j^τ needing its own estimate of $\hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})$
  $\rightsquigarrow$ With $|P|!$ permutations and n data points, the number of such estimates grows rapidly, making marginalization costly
- **Solution: Sampling-based approximation**
  Instead of computing $|P|! \cdot n$ terms, we approximate using M random samples of permutations τ and data points

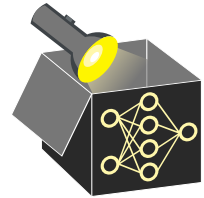
ESTIMATION: A PRACTICAL PROBLEM



- **Exact computation is infeasible for many features:**
  For $|P| = 10$, the number of permutations is $10! \approx 3.6$ million
  $\rightsquigarrow$ Complexity grows factorially with feature count
- **Additional challenge: Estimating marginal predictions (PD functions)**
  Each permutation τ defines a coalition S_j^τ needing its own estimate of $\hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})$
  $\rightsquigarrow$ With $|P|!$ permutations and n data points, the number of such estimates grows rapidly, making marginalization costly
- **Solution: Sampling-based approximation**
  Instead of computing $|P|! \cdot n$ terms, we approximate using M random samples of permutations τ and data points
- **Tradeoff: Accuracy vs. Efficiency**
  Larger M improves Shapley approximation
  $\rightsquigarrow$ Higher cost, but better fidelity to the exact value

APPROXIMATION ALGORITHM

Strumbelj et al. (2014)

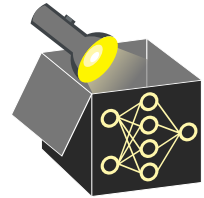


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:

APPROXIMATION ALGORITHM

Strumbelj et al. (2014)

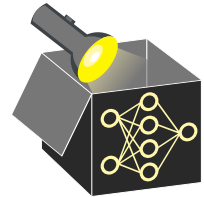


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
    1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices

APPROXIMATION ALGORITHM

Strumbelj et al. (2014)

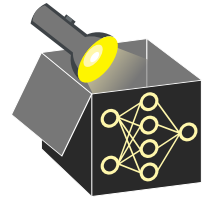


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^{\tau}$ be the set of features preceding j in τ

APPROXIMATION ALGORITHM

Strumbelj et al. (2014)

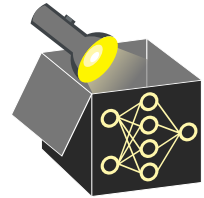


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
    1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
    2. Let coalition $S_m := S_j^{\tau}$ be the set of features preceding j in τ
    3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$

APPROXIMATION ALGORITHM

Strumbelj et al. (2014)

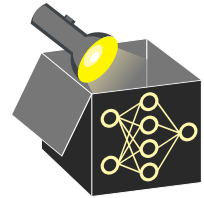


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^{\tau}$ be the set of features preceding j in τ
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$
   4. Construct two hybrid observations by combining values from $\mathbf{x}$ and $\mathbf{z}^{(m)}$:

APPROXIMATION ALGORITHM

▸ Strumbelj et al. (2014)

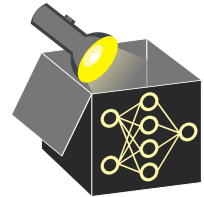


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^\tau$ be the set of features preceding j in τ
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$
   4. Construct two hybrid observations by combining values from $\mathbf{x}$ and $\mathbf{z}^{(m)}$:
      - $\mathbf{x}_{+j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, x_j, z^{(m)}_{\tau^{(|S_m|+2)}}, \ldots, z^{(m)}_{\tau^{(p)}})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m \cup \{j\}}$ (features in $S_m \cup \{j\}$ from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$

APPROXIMATION ALGORITHM

▸ Strumbelj et al. (2014)

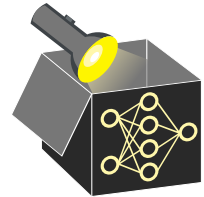


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^\tau$ be the set of features preceding j in τ
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$
   4. Construct two hybrid observations by combining values from $\mathbf{x}$ and $\mathbf{z}^{(m)}$:
      - $\mathbf{x}_{+j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, x_j, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m \cup \{j\}}$ (features in $S_m \cup \{j\}$ from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
      - $\mathbf{x}_{-j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, z_j^{(m)}, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m}$ (features in S_m excl. x_j from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$

APPROXIMATION ALGORITHM

▸ Strumbelj et al. (2014)

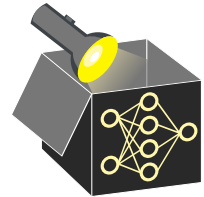


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^\tau$ be the set of features preceding j in τ
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$
   4. Construct two hybrid observations by combining values from $\mathbf{x}$ and $\mathbf{z}^{(m)}$:
      - $\mathbf{x}_{+j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, x_j, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m \cup \{j\}}$ (features in $S_m \cup \{j\}$ from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
      - $\mathbf{x}_{-j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, z_j^{(m)}, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m}$ (features in S_m excl. x_j from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
   5. Compute marginal contribution $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$

APPROXIMATION ALGORITHM

▸ Strumbelj et al. (2014)

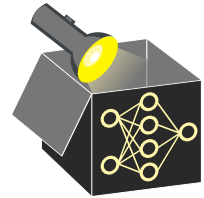


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^\tau$ be the set of features preceding j in τ
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$
   4. Construct two hybrid observations by combining values from $\mathbf{x}$ and $\mathbf{z}^{(m)}$:
      - $\mathbf{x}_{+j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, x_j, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m \cup \{j\}}$ (features in $S_m \cup \{j\}$ from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
      - $\mathbf{x}_{-j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, z_j^{(m)}, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m}$ (features in S_m excl. x_j from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
   5. Compute marginal contribution $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$
2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \Delta(j, S_m)$

APPROXIMATION ALGORITHM

▸ Strumbelj et al. (2014)

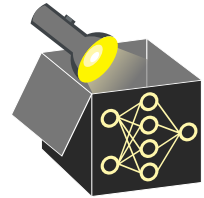


Estimate ϕ_j for feature j and model $\hat{f}$ trained on data $\mathcal{D}$ using M random samples:

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permutation $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^\tau$ be the set of features preceding j in τ
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$
   4. Construct two hybrid observations by combining values from $\mathbf{x}$ and $\mathbf{z}^{(m)}$:
      - $\mathbf{x}_{+j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, x_j, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m \cup \{j\}}$ (features in $S_m \cup \{j\}$ from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
      - $\mathbf{x}_{-j}^{(m)} = (x_{\tau^{(1)}}, \ldots, x_{\tau^{(|S_m|)}}, z_j^{(m)}, z_{\tau^{(|S_m|+2)}}^{(m)}, \ldots, z_{\tau^{(p)}}^{(m)})$
        $\rightsquigarrow$ includes $\mathbf{x}_{S_m}$ (features in S_m excl. x_j from $\mathbf{x}$), rest from $\mathbf{z}^{(m)}$
   5. Compute marginal contribution $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$
2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \Delta(j, S_m)$

$\rightsquigarrow$ Over M iterations, the PD functions $\hat{f}_{S_m}(\mathbf{x}_{S_m})$ and $\hat{f}_{S_m \cup \{j\}}(\mathbf{x}_{S_m \cup \{j\}})$ are approximated by $\hat{f}(\mathbf{x}_{-j}^{(m)})$ and $\hat{f}(\mathbf{x}_{+j}^{(m)})$, where features not in the coalition (to be marginalized) are substituted with values from the random data point $\mathbf{z}^{(m)}$

SHAPLEY VALUE APPROXIMATION - ILLUSTRATION



Definition

$\mathbf{x}$: obs. of interest

$\mathbf{x}$ with feature values in $\mathbf{x}_{S_m}$ (other are replaced)

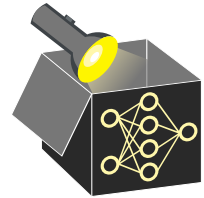
$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]$$

$\mathbf{x}$ with feature values in $\mathbf{x}_{S_m \cup \{j\}}$

	Temperature	Humidity	Windspeed	Year
x	10.66	56	11	2012
x_{+j}	10.66	56	$random: z_{windspeed}^{(m)}$	2012
x_{-j}	10.66	56	$random: z_{windspeed}^{(m)}$	$random: z_{year}^{(m)}$

j

SHAPLEY VALUE APPROXIMATION - ILLUSTRATION



Definition

Contribution of feature j to coalition S_m

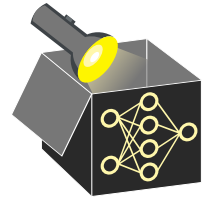
$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\underbrace{\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]}_{:=\,\Delta(j,\,S_m)}$$

- $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$ is marginal contribution of feature j to coalition S_m
- Here: Feature *year* contributes +700 bike rentals if it joins coalition $S_m = \{temp, hum\}$

	Temperature	Humidity	Windspeed	Year	Count
x	10.66	56	11	2012	
x_{+j}	10.66	56	$random: z_{windspeed}^{(m)}$	2012	5600
x_{-j}	10.66	56	$random: z_{windspeed}^{(m)}$	$random: z_{year}^{(m)}$	4900

j $\hat{f}$ $\Delta(j, S_m)$ marginal contribution: 700

SHAPLEY VALUE APPROXIMATION - ILLUSTRATION

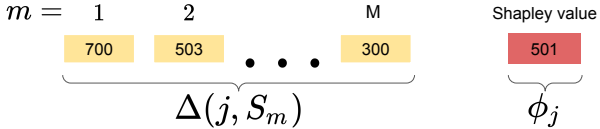


Definition

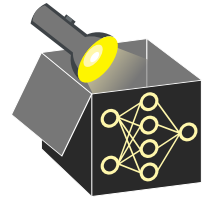
average the contributions of feature j

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]$$

- Compute marginal contribution of feature j towards the prediction across all randomly drawn feature coalitions $S_1, \ldots, S_m$
- Average all M marginal contributions of feature j
- Shapley value ϕ_j is the payout of feature j, i.e., how much feature *year* contributed to the overall prediction in bicycle counts of a specific observation $\mathbf{x}$



REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS



We adapt the classic Shapley axioms to the setting of model predictions:

- **Efficiency**: Sum of Shapley values adds up to the centered prediction:

$$\sum_{j=1}^{p} \phi_j(\mathbf{x}) = \hat{f}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}[\hat{f}(\mathbf{x})]$$

  $\rightsquigarrow$ All predictive contribution is fully distributed among features

- **Symmetry**: Identical contributors receive equal value:

$$\hat{f}_{S\cup\{j\}}(\mathbf{x}_{S\cup\{j\}}) = \hat{f}_{S\cup\{k\}}(\mathbf{x}_{S\cup\{k\}})\ \forall S \subseteq P \setminus \{j,k\} \Rightarrow \phi_j = \phi_k$$

  $\rightsquigarrow$ Interaction effects are shared equitably

- **Dummy (Null Player)**: Irrelevant features receive zero attribution:

$$\hat{f}_{S\cup\{j\}}(\mathbf{x}_{S\cup\{j\}}) = \hat{f}_S(\mathbf{x}_S)\ \forall S \subseteq P \Rightarrow \phi_j = 0$$

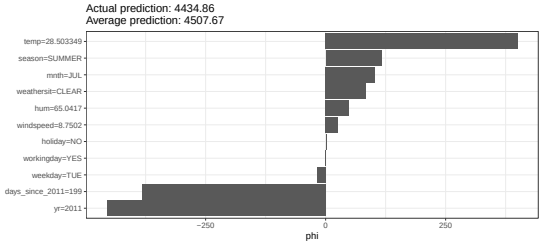
  $\rightsquigarrow$ Shapley value is zero for unused features (e.g., trees or LASSO)

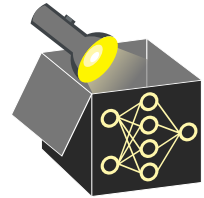
- **Additivity**: Attributions are additive across models:

$$\phi_j(v_1 + v_2) = \phi_j(v_1) + \phi_j(v_2)$$

  $\rightsquigarrow$ Enables combining Shapley values for model ensembles

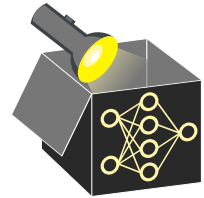
BIKE SHARING DATASET





- Shapley decomposition for a single prediction in bike sharing dataset
- Model prediction: $\hat{f}(\mathbf{x}^{(200)}) = \mathbf{4434.86}$ vs. dataset average: $\mathbb{E}_{\mathbf{x}}[\hat{f}(\mathbf{x})] = \mathbf{4507.67}$
- Total feature attribution: $\sum_j \phi_j =$ **-72.81**
  $\rightsquigarrow$ Explain downward shift from mean prediction
- Temperature (with value 28.5°C) is the strongest positive contributor: $+400$
- Features $\mathrm{yr} = 2011$ and $\mathrm{days_since_2011} = 199$ strongly reduce prediction
  $\rightsquigarrow$ Model captures lower bike demand in 2011 compared to 2012

ADVANTAGES AND DISADVANTAGES



Advantages:

- **Strong theoretical foundation** from cooperative game theory
- **Fair attribution:** Prediction is additively distributed across features
  $\rightsquigarrow$ Easy to interpret for users
- **Contrastive explanations:** Quantify each feature's role in deviating from the average prediction

Disadvantages:

- **Computational cost:** Exact computation scales factorially with feature count
  $\rightsquigarrow$ Without sampling, all 2^p coalitions (or $p!$ permutations) must be evaluated
- **Issue with correlated features:** Shapley values may evaluate the model on feature combinations that do not occur in the real data